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Title

Database Design Assignment Part B– Spring 2024 Apple App Store Case Study

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ASSIGNMENT COVERSHEET

UTS: ENGINEERING & INFORMATION TECHNOLOGY

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NAME OF TUTOR	TUTORIAL GROUP	DUE DATE
Wesley	Thursday 7:00PM	15 September 2024

ASSESSMENT ITEM NUMBER & TITLE
Database Design Assignment Part B– Spring 2024 Apple App Store Case Study

I acknowledge that if AI or another nonrecoverable source was used to generate materials for background research and self-study in producing this assignment, I have checked and verified the accuracy and integrity of the information used.

I confirm that I have read, understood and followed the guidelines for assignment submission and presentation on page 2 of this cover sheet.

I confirm that I have read, understood and followed the advice in the Subject Outline about assessment requirements.

I understand that if this assignment is submitted after the due date it may incur a penalty for lateness unless I have previously had an extension of time approved and have attached the written confirmation of this extension.

Declaration of originality: The work contained in this assignment, other than that specifically attributed to another source, is that of the author(s) and has not been previously submitted for assessment. I have rewritten any material provided by AI or other nonrecoverable sources and where appropriate acknowledged their contribution. I understand that, should this declaration be found to be false, disciplinary action could be taken and penalties imposed in accordance with University policy and rules. In the statement below, I have indicated the extent to which I have collaborated with others, whom I have named.

No content generated by AI technologies or other sources has been presented as my own work.

Statement of collaboration:

I would like to express my willingness to contribute to the Database Design Assignment Part B– Spring 2024 as a collaborator and equally take responsibility for completing my own part of the assignment.

Signature of student(s) Ran Vyvyan, Abdulaziz Almehmadi, Christopher Luu Date 15 September 2024

Database Design Assignment Part B

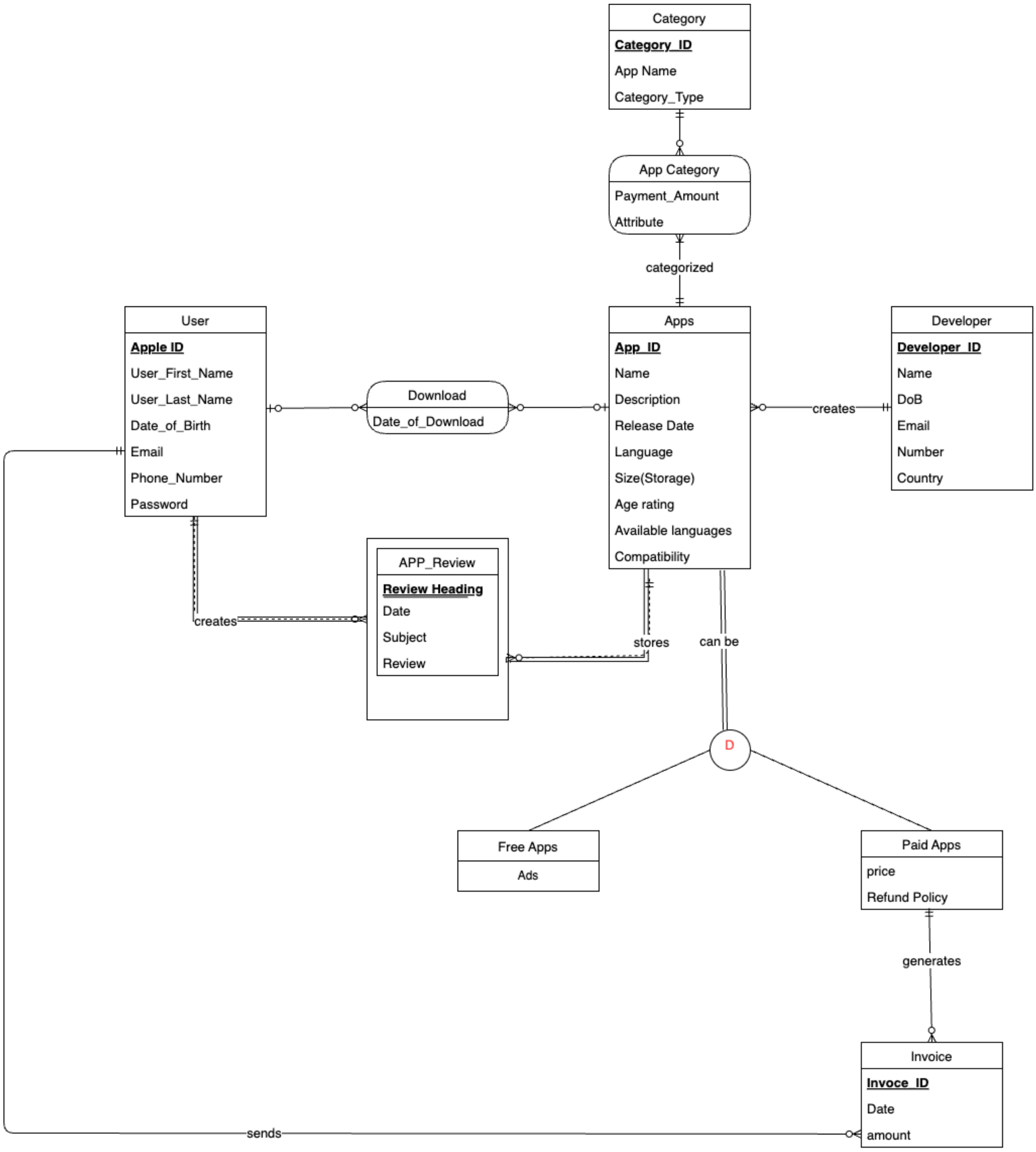
1. Overview of the case study in at most five sentences (From part A)

Apple App Store is an app marketplace that is compatible with the iPhone operating system and is exclusively used by users with an Apple ID. The system is required to collect information such as full name, email, phone number and date of birth in order to enable the app store service. The user can navigate to discover apps by using the search function or selecting one of the categories on the app store. Downloading the app might require providing further information depending on the app's cost and requirements.

2. Revised Business rules and Assumptions

1. A user can download/purchase many apps from the store but each download belongs to one user
2. A user must have an Apple ID to access the app store and user information (full name, date of birth, email address, phone number, and password) is required to create the ID.
3. Developers must also have a developer ID.
4. Users can provide one review per application they purchased and leaving a review is not mandatory.
5. Apps must display the developer's name and application size, as well as the app's version history.
6. Apps must display age calcification, size, descriptions, available languages, and the compatibility of the app with different devices and iOS versions.
7. Payments must be processed according to the user's billing address, including street, city, postal code, and country.
8. The price of the app and refund policy details must be clearly stated in the app store listing.
9. The free apps are labelled with a “Get” button.
10. Apps can belong to multiple categories.
11. Apps are identified with their unique ID.
12. Developers can develop multiple apps, and each app must belong to one developer.
13. Every invoice is identified by a unique Invoice_ID.
14. Each developer must provide their name, date of birth, email, contact number, and country.
15. Free apps contain attributes specific to free offerings (such as ad support or limited features).
16. When users purchase paid apps, an invoice is generated in the system.
17. Each category can have many apps
18. Apps can have many users.
19. Every invoice is associated with a single user, but each user can generate many invoices.
20. Each app can have multiple reviews submitted by various users.
21. Every review is associated with a specific app, and each review is for a single app only.
22. Each category has its unique category ID.

*We have added multiple business rules and modified the existing ones in Part A as we discovered many BRs needed for our ERD.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Email
Phone_Number
Password
Download
Date_of_Download
Apps
App_ID
Name
Description
Release Date
Language
Size(Storage)
Age rating
Available languages
Compatibility
creates
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Developer_ID
Name
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Email
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Country
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APP_Review
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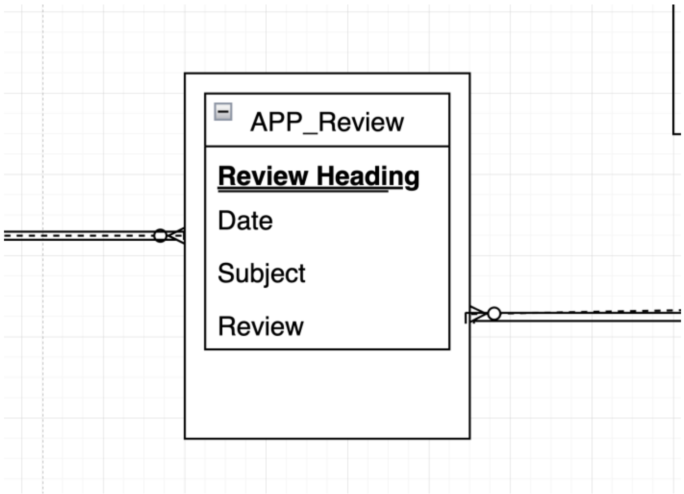
3. ERD



This is just to show a close-up of the diagram. Dash lines were just used to show the cardinalities here.

4. **Justifications** of the ERD based on the business rules and/or assumptions.

Entities

1. Business rules related to entity User:
BR2: A user must have an Apple ID to access the app store and user information (full name, date of birth, email address, phone number, and password) is required to create the ID.
- Primary Key (PK): Apple_ID

2. Business rules related to entity Apps:
BR5: Apps must display age calcification, size, descriptions, available languages, and the compatibility of the app with different devices and iOS versions.
BR11: Apps are identified with their unique ID.
- Primary Key (PK): App_ID

3. Business rules related to entity App Review:
BR4: Users can provide one review per application they purchased and leaving a review is not mandatory.
BR11: Apps are identified with their unique ID.
- Primary Key (PK): Foreign Key, APP_ID referencing APP entity
  Review Heading (on the diagram)
  Foreign Key, apple_Id referencing user entity

4. Business rules related to entity Developers:
BR3: Developers must also have a Developer ID.
BR14: Each developer must provide their name, date of birth, email, contact number, and country.
- Primary Key (PK): Developer_ID

5. Business rules related to entity Free Apps.
BR9: The free apps are labelled with a “Get” button.
BR15: Free apps contain attributes specific to free offerings (such as ad support or limited features).
- Primary Key (PK): Inferred but not explicitly defined; could use App_ID from Apps.

6. Business rules related to entity Paid Apps:
BR8: The price of the app and refund policy details must be clearly stated in the app store listing.
- Primary Key (PK): Inferred to be App_ID (shared with Apps)

7. Business rules related to entity Invoice:
BR7: Payments must be processed according to the user's billing address, including street, city, postal code, and country.
BR13: Every invoice is identified by a unique Invoice_ID.
BR16: When users purchase paid apps, an invoice is generated in the system.

- Primary Key (PK): Invoice_ID

8. Business rules related to entity Category:
BR10: Apps can belong to multiple categories.
BR17: Each category can have many apps
BR22: Each category has its unique category ID.

- Primary Key (PK): Category_ID

9. Business rules related to entity Download:
BR1: A user can download/purchase many apps from the store but each download belongs to one user.

- Composite Primary Key (PK): Foreign Key, Apple_ID from User, Foreign Key, App_ID from Apps

10. Business rules related to entity App Category:
BE10: Apps can belong to multiple categories.
BR11: Apps are identified with their unique ID.
BR17: Each category can have many apps
BR22: Each category has its unique category ID.

- Composite Primary Key (PK): Foreign Key, App_ID from Apps, Foreign Key, Category_ID from Category

Relationships and Cardinalities

1. Business Rules Related to the relationship between Apps to Developers and its cardinalities
BR12: Developers can develop multiple apps, and each app must belong to one developer.
(1 to many)
BR11: Apps are identified with their unique ID.

2. Business Rules Related to the relationship between Apps to Categories and its cardinalities
BR10: Apps can belong to multiple categories.(1 to many)
BR17: Each category can have many apps. (0 to many)
BR11: Apps are identified with their unique ID.
BR22: Each category has its unique category ID.

These entities have a many-to-many relationship. An associative entity (App Category) is involved.
The PK of the App Category: App_ID, Category_ID
TheFK of the App Category: App_ID from Apps, Category_ID from Category

3. Business Rules Related to the relationship between Apps to Users:
BR1: A user can download/purchase many apps from the store but each download belongs to one user.
(1 to many)
BR18: Apps can have many users (0 to many)

These entities have a many-to-many relationship. An associative entity (Download) is involved.
The PK of the Download: App_ID, Apple_ID
TheFK of the Download App_ID from Apps, Apple_ID from User

4. Business Rules Related to the relationship between Apps to Paid Apps:

BR8: The price of the app and refund policy details must be clearly stated in the app store listing.(Disjoint subtype of supertype - Apps)

5. Business Rules Related to the relationship between Apps to Free Apps:
BR9: The free apps are labelled with a “Get” button.
BR15: Free apps contain attributes specific to free offerings (such as ad support or limited features). (Disjoint subtype of supertype - Apps)

6. Business Rules Related to the relationship between User to Invoice:
BR19: Every invoice is associated with a single user, but each user can generate many invoices (1 to many).

7. Business Rules Related to the relationship between Apps to review:
BR4: Users can provide one review per application they purchased and leaving a review is not mandatory.
BR20: Each app can have multiple reviews submitted by various users. (0 to many)
BR21: Every review is associated with a specific app, and each review is for a single app only.

8. Business Rules Related to the relationship between Paid Apps to Invoice:
BR19: Every invoice is associated with a single user, but each user can generate many invoices. (1 to many)
BR7: Payments must be processed according to the user's billing address, including street, city, postal code, and country.
BR13: Every invoice is identified by a unique Invoice_ID.
BR16: When users purchase paid apps, an invoice is generated in the system.

9. Business Rules Related to the relationship between User to App review:
BR4: Users can provide one review per application they purchased and leaving a review is not mandatory. (0 to many)
BR21: Every review is associated with a specific app, and each review is for a single app only. (1 to many)
BR20: Each app can have multiple reviews submitted by various users.
BR2: A user must have an Apple ID to access the app store and user information (full name, date of birth, email address, phone number, and password) is required to create the ID.
BR11: Apps are identified with their unique ID.